

The Business Analyst Lab

Software Testing Life Cycle (STLC)

Notes for Business Analysts and Testers

What is STLC?

STLC (Software Testing Life Cycle) is a **step-by-step process** that defines all the activities carried out during **software testing** — from the beginning of the project to the release of the final product.

It helps ensure that **quality is maintained**, **defects are identified early**, and **software meets the business requirements**.

Why STLC is Important

- Ensures the **product meets customer expectations**.
- Helps to **detect and fix defects early**, saving cost and time.
- Improves **testing efficiency and coverage**.
- Provides **clear structure and process** to follow in testing.

Phases of STLC

STLC has **6 main phases**, as shown below:

1. **Requirement Analysis**
2. **Test Planning**
3. **Test Case Development**
4. **Test Environment Setup**
5. **Test Execution**
6. **Test Closure**

Let's understand each phase in detail 🖱️

1. Requirement Analysis

In this phase, the testing team studies and understands the **requirements document (BRD/SRS)** to identify what needs to be tested.

Key Activities:

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- Understand the **functional and non-functional requirements**.
- Identify **testable requirements**.
- Find any **missing or unclear requirements**.
- Prepare a **Requirement Traceability Matrix (RTM)**.

Deliverables:

- RTM
- Requirement analysis document

2. Test Planning

This is the phase where the **Test Manager or Senior Tester** creates a **test plan** — a roadmap of how testing will be done.

Key Activities:

- Define **scope and objectives** of testing.
- Identify **testing approach** (manual or automation).
- Estimate **effort, cost, and timelines**.
- Identify **test environment and tools**.
- Define **roles and responsibilities**.

Deliverables:

- Test Plan Document
- Effort Estimation Sheet

3. Test Case Development

In this phase, testers write **detailed test cases** that describe how to verify each functionality of the software.

Key Activities:

- Write **test cases** and **test data**.
- Review and finalize test cases.
- Create **RTM** mapping test cases with requirements.

Deliverables:

- Test Cases
- Test Data
- Updated RTM

4. Test Environment Setup

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A test environment is created that simulates the **actual production environment** where testing will happen.

Key Activities:

- Set up hardware, software, and network configurations.
- Install required applications and databases.
- Verify environment readiness (smoke testing).

Deliverables:

- Test Environment Setup Document
- Smoke Test Report

5. Test Execution

Now the actual testing begins. Testers execute the test cases on the application and record the results.

Key Activities:

- Execute test cases.
- Log **defects/bugs** in the defect tracking tool (like JIRA, Bugzilla, etc.).
- Re-test and perform **regression testing** after bugs are fixed.
- Update test results in RTM.

Deliverables:

- Test Execution Report
- Defect Report
- Updated RTM

6. Test Closure

This is the final phase of the STLC. After testing is completed, the team evaluates the testing process and shares the summary.

Key Activities:

- Ensure **all test cases** are executed and closed.
- Analyze **defect density and test coverage**.
- Prepare **Test Summary Report**.
- Identify **lessons learned** and **best practices** for future projects.

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Deliverables:

- Test Summary Report
- Test Closure Report
- Lessons Learned Document

Summary Table

Phase	Main Activities	Key Deliverables
Requirement Analysis	Study requirements, prepare RTM	RTM
Test Planning	Define scope, approach, timelines	Test Plan
Test Case Development	Write and review test cases	Test Cases, Test Data
Test Environment Setup	Set up environment	Environment Setup Doc
Test Execution	Run test cases, report bugs	Defect Report, Test Execution Report
Test Closure	Analyze results, prepare summary	Test Closure Report

Benefits of STLC

- Improves **product quality**.
- Provides **structured and systematic testing**.
- Saves **time and cost** by catching bugs early.
- Increases **transparency** and **accountability** in testing.

Real-Life Example

Example:

Imagine you are testing an **online shopping website**.

- During **Requirement Analysis**, you identify that users should be able to log in, search products, and make payments.
- In **Test Case Development**, you write test cases like “Check if user can log in with valid credentials.”
- In **Test Execution**, you run these cases and report if any functionality fails (e.g., payment gateway error).
- Finally, in **Test Closure**, you summarize how many test cases passed or failed and what defects were found.

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Common Tools Used in STLC

- **Test Management:** JIRA, TestRail, Zephyr
- **Automation:** Selenium, UFT, Katalon
- **Defect Tracking:** Bugzilla, JIRA
- **Reporting:** Excel, Power BI, or Test Summary Reports